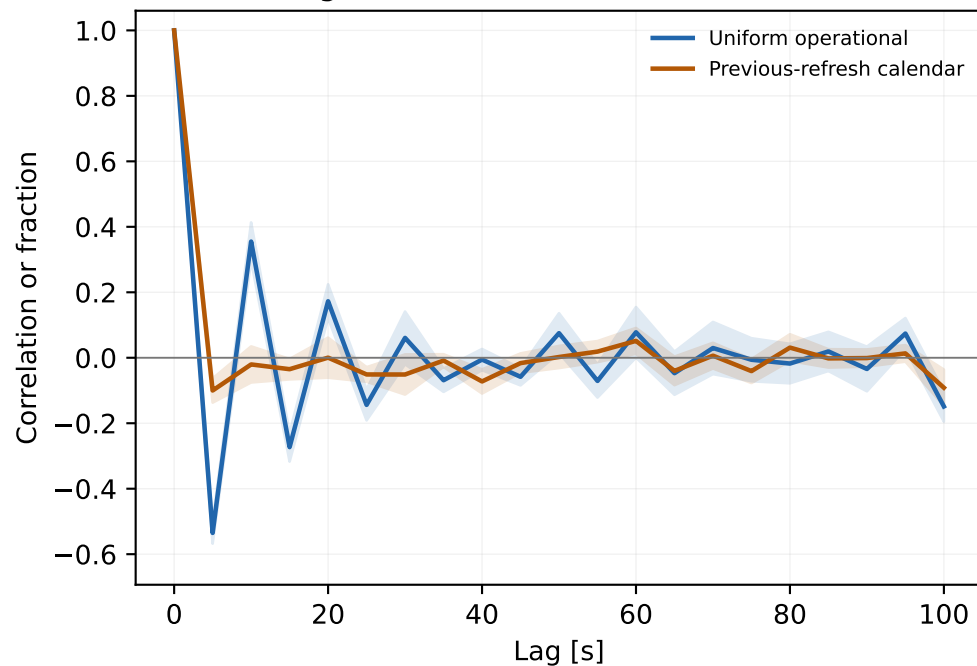
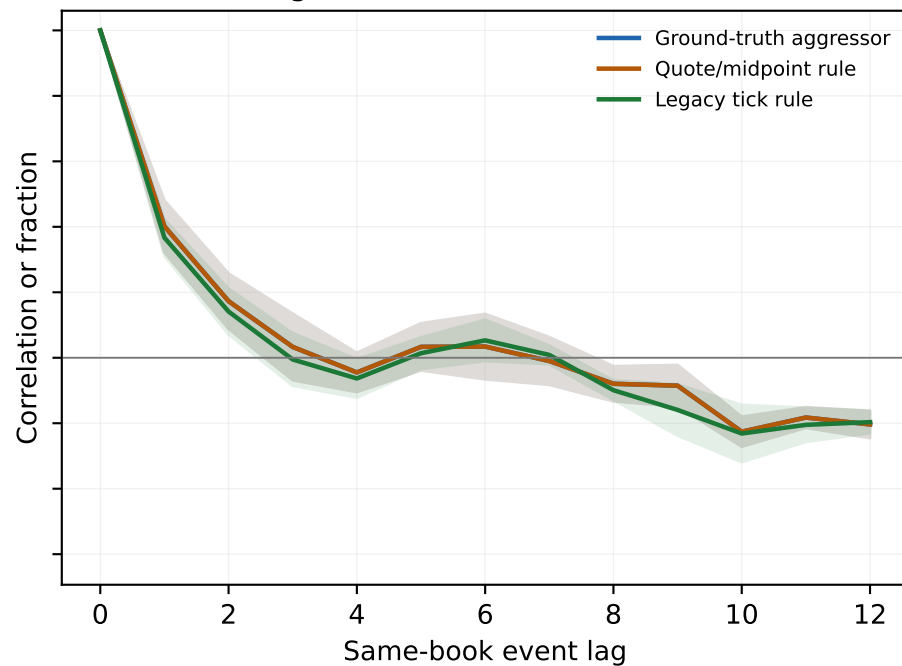


Mid-price and trade-sign dependence: event time, operational time, and explicit subordination

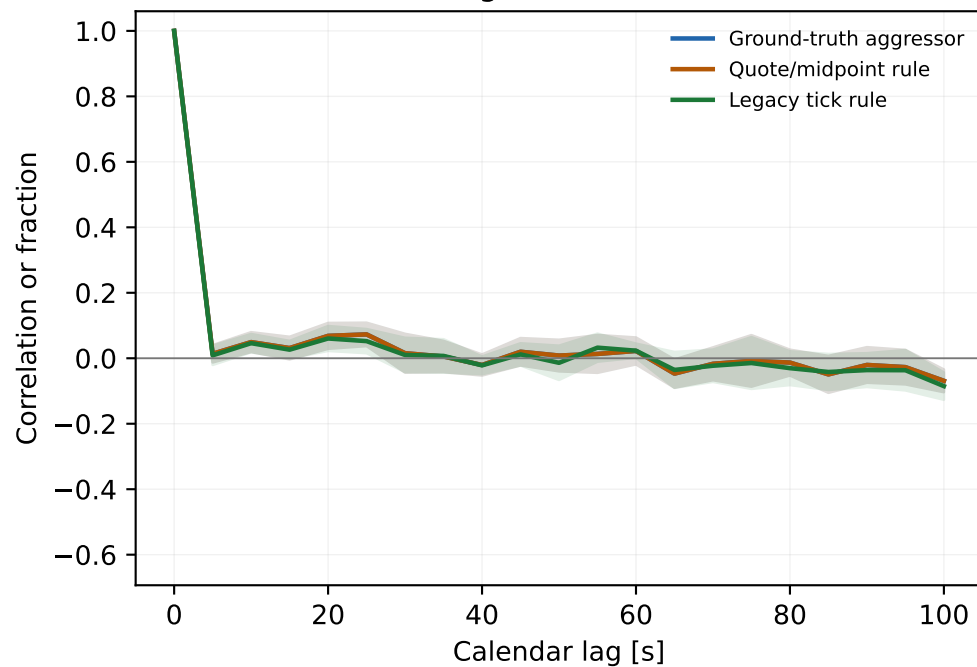
Log-mid increment autocorrelation



Trade-sign autocorrelation in event time



Subordinated signed-flow autocorrelation



Sign-convention agreement

